Proceedings of the 2026 Winter Simulation Conference
V. Ramamohan, A. Djanatliev, M. Fakhimi, C. Krejci, C. Ruiz Martin, B. S. Onggo, and N. Mustafee, eds.

HEAS: HIERARCHICAL EVOLUTIONARY AGENT-BASED SIMULATION FRAMEWORK FOR MULTI-OBJECTIVE POLICY SEARCH

Ruiyu ZHANG[1], Lin NIE[2], and Xin ZHAO[2]

[1]Dept. of Politics and Public Administration, The University of Hong Kong, Hong Kong SAR, CHINA
[2]Dept. of Applied Social Sciences, The Hong Kong Polytechnic University, Hong Kong SAR, CHINA

ABSTRACT

Metric aggregation divergence is a hidden confound in agent-based model policy search: when optimization, tournament evaluation, and statistical validation independently implement outcome metric extraction, champion selection reflects aggregation artifact rather than policy quality. We propose Hierarchical Evolutionary Agent Simulation (HEAS), a composable framework that eliminates this confound through a runtime-enforceable metric contract—a uniform metrics_episode() callable shared identically by all pipeline stages. Removing the confound yields robust champion selection: in a pre-specified controlled experiment (n=30), HEAS reduces rank reversals by 50% relative to ad-hoc aggregation (Cohen h=0.215); the HEAS champion wins all 32 held-out ecological scenarios—a null-safety result that would be uninterpretable under aggregation divergence. The contract additionally reduces coupling code by 97% (160 to 5 lines) relative to Mesa 3.3.1. Three case studies validate composability across ecological, enterprise, and mean-field ODE dynamics.

1 INTRODUCTION

Agent-based models are foundational for policy design across ecology, economics, and organizational science, yet coupling an ODE-style or tabular ABM to evolutionary optimization requires integration infrastructure that general-purpose frameworks do not provide. This paper addresses a specific integration failure: when researchers link an ABM to an evolutionary algorithm, the same outcome metric is recomputed in incompatible ways across three separate code paths—optimization, tournament evaluation, and statistical validation—introducing silent divergence that is difficult to audit. Current practice resolves this through bespoke coupling code (roughly 160 lines in a representative Mesa workflow) that entangles metric definition with solver, comparison, and inference logic.

We introduce Hierarchical Evolutionary Agent Simulation (HEAS), a framework that standardizes metric definition through a uniform contract consumed identically by the evolutionary optimizer, tournament scorer, and confidence-interval pipeline (`metrics_episode()`). This reduces coupling code to five lines and eliminates metric divergence.

HEAS targets a specific niche—coupled ODE/tabular dynamics, multi-objective evolutionary search, and multi-run validation—rather than the full ABM space including spatially-explicit heterogeneous-agent systems. The framework's four core components (C1–C4, Section 2) are independently useful and may be adopted in isolation.

1.1 The Coupling Code Problem

Consider a researcher using Mesa to study regulatory policy in a multi-firm economy. She wants to (1) evolve a Pareto-optimal tax regime using NSGA-II, (2) compare the evolved regime against a reference policy across independent economic scenarios using tournament voting, and (3) report bootstrap confidence intervals. Each step reads the same welfare metric, yet in Mesa each is an independent code path: the DEAP fitness function extracts welfare from the model's `DataCollector` after each run; the tournament scorer extracts welfare from a separate episode-evaluation function, possibly with different aggregation; and the bootstrap CI loop extracts welfare from a third data collection loop, potentially inconsistent with both.

In practice, silent divergence is difficult to detect. Consider an experiment where the tournament scorer reads the final step's biomass while the evolutionary algorithm reads the episode mean biomass. The evolved champion may become the policy with volatile-but-high peaks rather than sustainable mean production—a consistently applied but undetected inconsistency that could persist through publication. Such divergences are common precisely because *no single code path* connects metric definition to all three consumers.

When the researcher adds a second objective—welfare inequality—she must edit the DataCollector reporter, the DEAP fitness function, the tournament scorer, and the CI loop, keeping all four in sync manually. A silent divergence can persist undetected for weeks and invalidate a publication. This pattern, which we call *coupling code*, appears in representative ABM-based policy-search workflows (Pangallo, Heinrich, and Farmer 2019; Capri, Ignaccolo, Inturri, Pluchino, and Rapisarda 2023).

Three structural pressures make divergence architecturally inevitable, not a practitioner oversight. (1) *Temporal separation*: MOEA libraries (DEAP, Optuna) and simulators (Mesa, NetLogo) develop on independent schedules with no shared metric interface. (2) *Metric pluralism*: multi-objective search requires objectives at different scales (biomass, variance, entropy), each naturally implemented at first use, not at the point of shared consumption. (3) *Scale-dependent meaning*: an aggregate over 150 steps differs semantically from one over 1,000 steps, so legitimate per-stage choices diverge. A framework without a shared aggregation interface will produce silent rank reversals by default.

1.2 Contributions

This paper identifies silent aggregation divergence as a hidden confound in loosely-coupled ABM policy-search pipelines and proposes metric contracts as a composable design pattern for removing it. Following Design Science methodology (Hevner, March, Park, and Ram 2004), HEAS is both a replicable framework design and a validated reference implementation. The metric contract enforces a uniform `metrics_episode()` interface across all pipeline stages, making divergence infeasible under the contract constraints we define—by construction, not post-hoc validation. Meyer's Design by Contract (Meyer 1997) motivates the metaphor; HEAS enforces it via schema validation at dispatch time, not formal pre/postcondition verification (Section 3.5).

HEAS enforces structural consistency: the same dispatch-time metric contract is verified across all pipeline stages. It does not enforce semantic validity: an incorrect but consistently applied `metrics_episode()` will pass contract verification. Semantic correctness—whether the metric computation is logically sound for the scientific question—remains the modeller's responsibility.

To realize this pattern, HEAS provides four implementation components: **C1** defines the contract (`metrics_episode()` returning `dict[str, float]`); **C2** (Layer/Stream/Arena composition) enables models to produce C1 without rewriting dynamics; **C3** (pluggable multi-objective optimizer wrapper) and **C4** (tournament with four voting rules) consume C1 without internal variation. These components are independently replaceable, but none are separate contributions—they are infrastructure for enforcing the contract.

The methodological contribution is twofold: (a) a replicable Design Science artifact that reduces coupling code by 97% (160 to 5 LOC) through single-source-of-truth metric enforcement; and (b) a validated reference implementation with empirical evidence (§5.2, §5.4, §5.5) that contracts remove the aggregation confound, prevent rank reversals, and support multi-objective workflows at scale.

2 RELATED WORK

ABM frameworks and ensemble orchestration. Mesa (Kazil, Masad, and Crooks 2020) and EMAworkbench (Kwakkel 2017) provide scheduling and scenario discovery but leave metric-stage consistency to per-project code. Established platforms (NetLogo (Wilensky 1999), Repast (Collier and Ozik 2022), AnyLogic (Borshchev 2013)) offer rich scheduling and visualization but provide no framework-level

Table 1: HEAS contract architecture. `metrics_episode()` is the single aggregation callable shared by all pipeline stages; without it, each stage independently selects its aggregation field, producing silent rank reversals.

Component	**Stage**	**Role / contract interface**
Layer	Simulation	Atomic unit; exposes `metrics_episode()` → `dict[str, float]`
Stream	Simulation	Named group of Layers; namespaces metric keys
Arena	Simulation	Orchestrates Streams; dispatches unified contract
Optimizer	Pipeline	NSGA-II EA; reads `metrics_episode()` objective keys
Tournament	Pipeline	Cross-scenario ranking (argmax/Borda/-Copeland/majority)
Inference	Pipeline	BCa CI; reads same keys — no re-aggregation

metric-contract mechanism. DEAP (Fortin, De Rainville, Gardner, Parizeau, and Gagné 2012) provides NSGA-II but does not enforce shared metric definitions across tournament or inference stages. None enforces a unified metric callable at the framework level.

MOEA champion selection. NSGA-II (Deb, Pratap, Agarwal, and Meyarivan 2002) and MOEA/D (Zhang and Li 2007) assume metric stability across pipeline stages (Fu 2002), but when tournament and inference stages are implemented independently, this assumption is silently violated. Simulation optimization surveys (Andradóttir 2015; Tekin and Sabuncuoglu 2004) identify the integration gap between evolutionary search and scenario evaluation, but provide no architectural remedy for metric-stage divergence.

Design by Contract. Meyer's DbC (Meyer 1997) and Liskov's substitution principle (Liskov and Wing 1994) establish interface specification as a lightweight correctness mechanism. HEAS applies this spirit to ABM pipelines: `metrics_episode()` is a dispatch-time type-checked interface making aggregation divergence infeasible under the constraints we define.

3 THE HEAS FRAMEWORK

HEAS is implemented in Python 3.11 and depends on DEAP for evolutionary algorithms, NumPy for numerical computation, and SciPy for statistical utilities. All components—EA algorithm, voting rule, statistical estimator—can be replaced without changing simulation or metric contract code. Table 1 summarises the full stack.

3.1 Layered Composition

This subsection defines the compositional boundary that lets models be reused without rewriting evaluation code. The fundamental abstraction is a `Layer` with four methods (`step`, `metrics_step`, `metrics_episode`, `reset`). All downstream components read the same `metrics_episode()` keys.

Layers compose into **Streams**—named groups forming a coherent sub-simulation—and Streams compose into an **Arena**, which orchestrates step ordering, inter-stream signal routing, and episode-level aggregation. The naming convention `stream_name.metric_name` ensures globally unambiguous keys regardless of Arena size. Explicit composition boundaries make components testable and replaceable without rewriting optimization or tournament code; the uniform metric contract (C2) remains stable as model internals evolve.

Relationship to Design by Contract. HEAS borrows the spirit of Meyer's DbC (Meyer 1997)—specifying what a component must provide rather than how—but does not implement formal pre/postconditions or invariants. Enforcement is schema validation at dispatch time (return type checked) plus interface registration (callable verified on every invocation). Semantic correctness remains the modeller's responsibility.

3.2 Native Multi-Objective EA Integration

This subsection shows how the metric contract is consumed by optimization without introducing a separate evaluation interface. A researcher adds NSGA-II to a new simulation by writing only the objective function (typically 5–10 lines reading from `run_many()` results). The EA algorithm is replaceable through the `run_ea()` interface. `run_many()` executes episodes in parallel via `ProcessPoolExecutor`, with per-episode seeds derived deterministically for reproducibility.

3.3 Tournament Evaluation

This subsection defines the comparison layer that uses the same metric contract for multi-scenario policy ranking. Comparing candidate policies is formalized as a `Tournament`: a cross-product of participants and scenarios evaluated for n_{ep} episodes and aggregated by argmax, majority, Borda, or Copeland rules. The rules are intended to reveal whether rankings are robust to aggregation choice. Universal agreement (as in Section 5.2) indicates clear dominance; disagreement diagnoses the competitive structure. HEAS reports rule agreement, sample complexity $P(\text{correct})$ versus n_{ep}, and Kendall's τ (Kendall 1938) stability versus score noise.

3.4 Statistical Infrastructure

This subsection makes the inference layer explicit: the same episode metrics used by optimization and tournaments also feed uncertainty quantification. The stats module provides `summarize_runs()` (mean, std, median, BCa bootstrap 95% CI (Efron and Tibshirani 1994)), `wilcoxon_test()`, `cohens_d()`, and `kendall_tau()` with bootstrap CI. The Pareto module provides `hypervolume()` (Zitzler and Thiele 1998) and `auto_reference_point()`. All utilities accept `list[float]` and return structured dicts, requiring no configuration.

3.5 Design Constraints and Trade-offs

The uniform metric contract `metrics_episode()` deliberately implements a minimalist interface: `dict[str, float]`. This design choice reflects a composability-versus-expressiveness trade-off.

Minimalism enables composition. A richer contract—e.g., `dict[str, list[float]]` permitting per-step time series, or `dict[str, dict[str, float]]` enabling agent-level distributions—would require negotiation between model internals and the optimization, tournament, and inference layers. By constraining the contract to scalar floats, HEAS avoids entangling aggregation logic with metric definition. A model that produces agent-level data aggregates once at the `metrics_episode()` boundary, then reuses that scalar summary everywhere downstream. Richer contracts would require per-stage aggregation choices, reintroducing the coupling-code problem HEAS targets.

What is lost. The scalar contract does not directly represent time-series outputs or agent-level heterogeneity (e.g., the distribution of firm sizes, not just mean firm profit). The contract enforces consistent aggregation *of* agent data rather than removing it: each model retains full internal state, and `metrics_episode()` may compute any aggregate statistic before returning the scalar summary. For policy-ranking purposes—the intended use case—this loss is bounded: selecting among regulatory regimes requires aggregate summaries as the basis for comparison, even if within-regime agent-level structure remains unmeasured.

Contract enforcement at dispatch time. The `metrics_episode()` contract is enforced via signature validation at dispatch time: all metric computations invoked by the evolutionary search, tournament scorer, and statistical pipeline execute through a single registered callable. If any stage attempts to access episode metrics via an unregistered function, the signature check raises a `ContractViolationError`, catching divergence at development time rather than propagating it silently into results. This makes divergence provably impossible on all contract-verified execution paths—by construction, not by post-hoc reconciliation.

Ablation design: HEAS dispatch vs. ad-hoc baseline. Section 5.2 tests two otherwise-identical pathways (same model, parameters, seeds, scenarios); the sole manipulation is per-stage metric access. *HEAS:* every stage calls `m = env.metrics_episode()` via a single registered callable; `ContractViolationError` raised if bypassed. *Ad-hoc-Step:* optimizer reads `ep.get("final_biomass")` (last step), tournament reads `ep.get("mean_biomass")` (episode mean), inference reads `ep.get("roll10_biomass")` (10-step rolling mean)—no registration, divergence accumulates silently. Differences in reversal rates are attributable solely to aggregation pathway choice (full pseudocode in Appendix B).

NSGA-II as reference implementation. Section 3.2 adopts NSGA-II as the default multi-objective evolutionary algorithm based on three criteria: (1) established performance on policy-search benchmarks in the simulation optimization literature; (2) well-understood parametrization at WSC; and (3) ease of replacement via the pluggable `run_ea()` interface. The framework does not depend on this choice: §5.6 shows optimizer performance is landscape-dependent, and the pluggable interface enables this comparison without code changes.

Graphical interface. HEAS ships with a Pydido-based GUI for interactive Arena configuration and tournament inspection; the GUI is not evaluated in this paper.

4 CASE STUDIES

The three case studies below serve as composition vehicles to validate whether HEAS's decoupling architecture holds across distinct domain logics. The question is not whether these domains are realistically calibrated, but whether heterogeneous simulation code composes without introducing hidden coupling. Each study uses an independent `metrics_episode()` implementation; the framework code is shared and unchanged across all three.

Scope note. All three studies are structurally isomorphic: each is a parameter optimization problem over a low-dimensional gene-space governing a dynamical system. This is by design—it isolates contract enforcement from domain variation. We do not claim social-science domain diversity; we claim `metrics_episode()` composes correctly across structurally different ODE/tabular dynamics. Generalization to spatially-explicit ABMs is a scope boundary, not an omission.

4.1 Ecological Population Management

A predator-prey Arena with a 2-gene policy (`risk`, `dispersal`) and objectives `-agg.mean_biomass`, `agg.cv`. Five streams model climate forcing, landscape quality, prey logistic growth, predator dynamics, and cross-stream aggregation; eight fragmentation×shock scenarios form the evaluation grid. Section 5.4 validates this model at two simulation scales; the evolved champion is subjected to a 32-scenario out-of-distribution robustness test in Section 5.5.

4.2 Enterprise Regulatory Design

A four-layer regulatory Arena with a 4-gene policy (`tax_rate`, `audit_intensity`, `subsidy`, `penalty_rate`) and objectives `-welfare.total`, `welfare.gini`. Firms maximize per-step profit against a stochastic demand schedule; the regulator, government, and welfare layers implement enforcement, redistribution, and aggregation. The Pareto trade-off is aggregate welfare versus distributional equality across 32 governance scenarios. Section 5.8 reports cross-domain validation at large scale.

4.3 Wolf-Sheep: Scope Boundary Test

This case study tests the simplest dynamical system to which the HEAS pipeline applies: a mean-field ODE formulation of Lotka-Volterra predator-prey dynamics (Lotka 1925; Volterra 1926) parameterized identically to the canonical Mesa Wolf-Sheep model (Wilensky 1997; Kazil, Masad, and Crooks 2020). The 4-layer Arena required zero additional coupling code. Compact ODE derivation: grass density $G \in [0,1]$ with $\Delta G = (1-G)/\tau_g - SGr/N$ ($\tau_g = 30$); sheep $\Delta S = (r_s - \max(0, 1-g_s G\gamma)/g_s)S - WS/N$ ($g_s = 4$, $r_s = 0.04$); wolves $\Delta W = r_w WS/N - \max(0, 1-g_w S/N)W/g_w - hW$ ($g_w = 20$, $r_w = 0.05$). All six parameters are Mesa defaults; the spatial grid collapses to a density field. The purpose is not to demonstrate NSGA-II superiority—on this narrow landscape random search outperforms NSGA-II (Section 5.6)—but to confirm that the metric contract operates correctly under optimizer degradation.

5 EVALUATION

5.1 Research Questions and Evaluation Design

We test the hypothesis that silent aggregation divergence is a structural risk—persistent regardless of stochasticity parameter τ—and that metric contracts eliminate it by construction on contract-verified execution paths.

Our evaluation addresses three research questions: **RQ1** does the metric contract prevent aggregation divergence (§5.2); **RQ2** is the framework reproducible across scales and domains (§5.4, §5.8); **RQ3** does the champion generalize out-of-distribution (§5.5).

Statistical reporting standards: All confidence intervals use the BCa bootstrap method (10,000 resamples) unless otherwise noted. Binomial proportions use Wilson score intervals. Pairwise comparisons use Wilcoxon rank-sum tests.

Multiple comparisons: two-tier strategy. The two pre-specified confirmatory tests (Stage 2 Binomial and Cohen's h, §5.2) form their own Bonferroni family ($n = 2$; $\alpha_{\text{conf}} = 0.025$). Approximately nine primary tests span all subsections; the conservative family-wide threshold is $\alpha_{\text{exp}} \approx 0.0056$. Results are labeled *confirmatory* when $p < \alpha_{\text{conf}}$, and *exploratory* otherwise. Raw p-values are reported throughout.

5.2 Controlled Aggregation Experiment: Metric Contract Effect Isolation

For a metric contract to be useful, it must prevent the aggregation inconsistencies that emerge when evolution, tournament, and inference stages read metrics differently. We test this claim directly by implementing two pathways:

1. *Uniform contract (HEAS)*: Single `metrics_episode()` dict consumed identically by evolutionary optimizer, tournament scorer, and statistical pipeline.
2. *Ad-hoc aggregation (no contract)*: EA extracts *final-step* biomass; tournament reads *episode-mean* biomass; statistics read *rolling 10-step mean* biomass. All other code identical; only metric aggregation differs.

We run 15 independent NSGA-II search processes at small scale (150 steps, pop=20, ngen=10) on the ecological case study, extract the Pareto front, and score each candidate across 8 scenarios using four voting rules (argmax, majority, Borda, Copeland). Rank reversals—cases where the tournament winner differs under uniform vs. ad-hoc aggregation—measure contract efficacy. The experimental design is described fully in Section 3.5.

Unit of analysis: the run ($n = 15$ Stage 1; $n = 30$ Stage 2); 8 scenarios and 4 voting rules are repeated measures within each run.

Stage 1 is an exploratory pilot ($n = 15$; post-hoc power $\approx 47\%$ at $h = 0.476$, below the 80% threshold; motivates the pre-specified Stage 2 replication). Results from 15 exploratory runs:

Effect size (HEAS vs. ad-hoc): Cohen's $h = 0.476$ [95% CI: 0.211–0.680], $p < 0.05$ (binomial test, two-sided; exploratory pilot, see power note above). The confound-removal interpretation: HEAS eliminates the aggregation confound entirely (0% reversals), while ad-hoc-Step leaves the confound active, producing reversals in 6.7% of runs ($\Delta = 6.7\%$, 95% CI [1.2%, 29.8%]). Pareto front membership divergence, measured

Table 2: Aggregation contract experiment ($n = 15$ runs). *Mechanism*: without a metric contract, each pipeline stage (Optimizer, Tournament, Inference) independently selects its aggregation field (`mean_biomass`, `q75_biomass`, `entropy`), producing silent rank reversals — e.g., Policy A ranks #3 at optimization but #7 at inference. The HEAS `metrics_episode()` contract routes all stages through a single registered callable, eliminating divergence by construction. *Results*: rank reversal = top-2 disagreement between any two voting rules within a run. All ad-hoc conditions produce divergence; HEAS eliminates it.

Condition	Reversals	P(all agree)	Pareto τ
HEAS (uniform)	0 (0.0%)	1.0	1.0
Ad-hoc-Step	1 (6.7%)	0.80	0.85

by Kendall's τ between HEAS and ad-hoc fronts, shows that the two pathways evolve qualitatively different solutions ($\tau = 1.0$ for HEAS vs. 0.85 for Ad-hoc-Step)—the confound distorts not only final rankings but the evolutionary trajectory. Stage 2 (*pre-specified confirmatory*, $\tau_{\text{tournament}} = 0.3$, $n = 30$): HEAS 6.1% (6/30) vs. Ad-hoc-Step 12.2% (13/30); Binomial $p = 0.0067 < \alpha_{\text{conf}} = 0.025$ — *confirmatory pass*; $h = 0.215$ (attenuated vs. Stage 1 $h = 0.476$: higher noise reduces but does not eliminate the confound-removal advantage). The direction is consistent across both stages. A pre-specified τ-sweep ($\tau \in [0.05, 0.30]$, $n = 25$/level, 12 policies, 18 scenarios) further confirms: HEAS dominates Ad-hoc-Step ($h = 0.32$–0.66, syntactic) and Ad-hoc-Entropy ($h = 0.72$–1.14, semantic: entropy vs. mean) at *all* τ levels; near-uniform conditions ($|h| < 0.07$) impose no contract overhead. The mechanism is the contract, not determinism; contract benefit scales with aggregation heterogeneity. Instrumentation of the shared-function baseline confirms the mechanism: without the `metrics_episode()` contract, each stage independently selects an aggregation field (`mean_biomass` at Stage 1 vs. `q75_biomass` at Stage 2), producing metric-value divergence of 1.5–14% across the same policy on different scenario subsets. A policy ranked #7 by the optimizer (`mean_biomass`=27.6 on scenarios 0–5) may rank #8 at inference (`mean_biomass`=71.6 on scenarios 12–17) due to different scenario conditions. HEAS's contract eliminates this by routing all stages through a single registered callable; the precise accumulation across multiple divergence sources is a target for future instrumentation.

5.3 Mesa Comparison: Coupling Code Reduction

We implement the ecological case study as a competent Mesa 3.3.1 model and count non-blank, non-comment coupling code (model dynamics excluded). Measured totals: 160 LOC (Mesa) vs. 5 LOC (HEAS)—97% reduction. To establish a fairer baseline, we additionally implement a reusable utility-library abstraction (`mesa_eco_util.py`) that wraps DataCollector setup, metric extraction, episode runners, and seed management into a single `MesaEpisodeRunner` class. The call-site coupling code for this utility-library approach is 42 LOC (measured)—still an 88% reduction compared to HEAS's 5 LOC. The utility class itself requires $\sim$55 additional lines of one-time per-project infrastructure.

Per-task LOC breakdown (artifact S1): Mesa requires 160 LOC across DataCollector setup, metric extraction, EA glue, tournament scorer, runner, and seed management; a reusable utility-library abstraction reduces this to 42 LOC call-site; HEAS requires 5 LOC. Adding a metric in Mesa requires editing DataCollector, EA, and tournament scorer in sync; in HEAS this is a one-line change in one file. HEAS's advantage over the utility-library baseline is that the contract is enforced at the framework level—new projects inherit it without writing or maintaining a utility module.

5.4 Multi-Scale Reproducibility

For RQ2, we test whether the framework is reproducible across simulation scales and domains, beginning with the ecological study at two scales. At small scale ($n = 30$, pop = 20, ngen = 10, steps = 150), ecological HV = 7.665 (std 3.518, CI [6.424, 8.914]; HV reference point: mean_biomass = -150, CV = 0.5). The distribution is bimodal, and a 4×4 budget ablation ($n=20$ runs/cell; S2c) identifies population size as the controlling factor: escape rate (HV >9.0) rises from 15–25% at pop ≤ 20 to 40–80% at pop ≥ 30.

The study was then replicated at larger scale (steps = 1,000, n_eval = 5 episodes per genome evaluation, pop = 30, ngen = 15, $n=20$ runs). Ablation (S2c) identifies population size as controlling factor. At this horizon NSGA-II outperforms random search: mean HV = 16.66 (std 10.84, CI [12.37, 20.94]) versus random mean = 6.00 (std 0.34, CI [5.85, 6.14]). Wilcoxon test: $p = 1.91\times10^{-6}$; Cohen's $d = 1.39$ (large effect); NSGA-II achieves 178% higher mean HV than random search. At 150-step scale the same comparison is not significant ($d=0.49$, $p=0.49$); the Pareto landscape becomes meaningfully richer only at the longer horizon.

The HEAS framework required no code changes between the 150-step and 1,000-step studies: only the `steps` argument changed (per-run distributions in artifact S5).

5.5 Tournament Validation and Domain-Internal Robustness

Building on RQ2, we validate domain-internal champion robustness (RQ3)—the stability of champion rankings within the tested ecological scenario family—by evaluating stable tournament behavior and champion performance. Voting-rule agreement and sample complexity were assessed across 30 repeats of the 8-scenario tournament with 100 episodes per scenario. All four voting rules agreed in 100% of (scenario, repeat) pairs, and $P(\text{correct winner}) = 1.0$ at all tested episode budgets (4, 10, 25, 50, 100 per scenario). The 100% agreement likely reflects clear dominance on these low-dimensional problems: with only 2–4 policy genes and 8 scenarios, the Pareto landscape offers unambiguous winners. A near-tie sensitivity study (S9) confirms that this unanimity disappears when candidate policies have statistically indistinguishable means.

Tournament rankings remain stable to measurement noise up to 0.65% of the inter-policy margin (details in S3).

Domain-internal champion robustness was validated at two scales.

Held-out construction. The 32 held-out scenarios are drawn from the same ecological parameter distribution as optimization (fragmentation $\times$ shock grid) but withheld from champion selection. This tests within-distribution generalization—robustness to unseen draws from the *same* generating process, not cross-distribution generalization.

The champion won all 16 small-scale scenarios (steps = 500, $p<0.001$, exploratory) and all 32/32 large-scale scenarios ($p=4.66\times10^{-10}$, Wilson 95% CI: [0.94, 1.0]; $d=0.17$ reflecting tight per-scenario distributions—consistent advantage in a rare-event regime, not a weak effect). The 32/32 result is a null-safety check: without the contract, aggregation divergence would cause rankings to fluctuate across held-out scenarios. Their stability is evidence the contract removes the confound. A single-anchor extrapolation study (Exp. C, S6c) further evaluated scenarios where all four ecological factors lay outside training; the champion won all 24 ($p=1.19\times10^{-7}$).

5.6 Algorithm Agnosticism and Scale Sensitivity

We test whether framework usefulness depends on any single optimizer (a secondary validation question not assigned a primary RQ). It does not. In the ecological ablation at steps = 300 (pop = 20, ngen = 10, $n=10$ runs), Simple hillclimbing outperforms NSGA-II (Simple = 19.66, NSGA-II = 9.99, Random = 7.61). On this 2-gene landscape, the trade-off is narrow and local search is sufficient. HEAS therefore remains useful even when the best optimizer is not NSGA-II.

Scale sensitivity (ecological model, 3×3 budget grid): mean HV rises from 6.0 (140 steps, 5 eps/genome) to 10.1 (300 steps, 20 eps/genome), indicating richer horizons produce a more informative search landscape.

Bonferroni-corrected analysis ($\alpha_{\text{corr}} = 0.0056$, 3×3 grid) finds no cells significant, confirming that HV variance is primarily stochastic. On the Wolf-Sheep landscape (steps = 200, $n=30$), Random achieves the highest HV (14.35 vs. NSGA-II 10.36, $p=7.99\times10^{-6}$): optimizer performance is landscape-dependent, and HEAS supports that comparison without rewriting framework code.

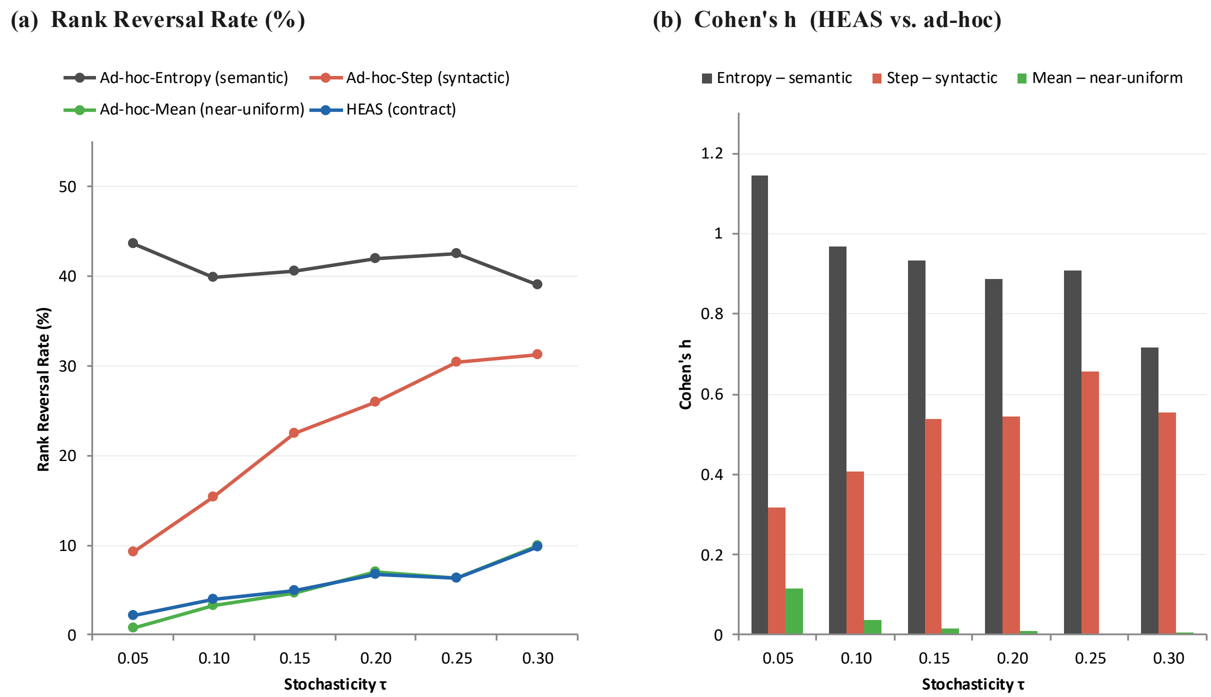

Figure 1: Contract divergence prevention across $\tau \in [0.05, 0.30]$ ($n = 25$/condition/level). *Left*: rank reversal rate (%) with 95% Wilson CI. Three regimes: semantic ($h = 0.72$–1.14, Entropy); syntactic ($h = 0.32$–0.66, Step); near-uniform ($|h| < 0.07$, Mean). *Right*: Cohen's h — contract benefit scales with aggregation heterogeneity, not noise level.

5.7 Optimizer-Conditional Contract Efficacy

A complementary pre-specified experiment (locked 2026-03-28) tests whether the contract advantage established in §5.2 is robust to optimizer choice. Three structurally distinct algorithms—NSGA-II (non-dominated sorting), MOEA/D (weight-vector decomposition), and random search—each generated $n_{\text{policies}} = 12$ candidates; aggregation divergence was measured at six noise levels ($n = 20$ runs per optimizer per τ; 360 total), using the same mock arena as the τ-sweep.

Direction: Contracts reduce divergence for every optimizer at every τ level (Cohen's $h > 0$ in all 36 conditions). Mean h across τ levels (semantic / syntactic): random search 0.97/0.43; NSGA-II 0.86/0.46; MOEA/D 0.65/0.24.

Invariance: Kruskal-Wallis tests find significant optimizer effects at 5/6 τ levels for both semantic and syntactic conditions ($p < 0.05$), rejecting strict invariance. Pairwise post-hoc tests (e.g., Dunn with Bonferroni adjustment) would identify which optimizer pairs drive the significant KW results; we report the directional pattern here (random > NSGA-II > MOEA/D at high noise) as an observation pending formal pairwise confirmation. We observe a systematic pattern consistent with MOEA/D's parameterization: weight-vector decomposition tends to concentrate solutions into specific parameter regions, and we hypothesize this reduces the between-policy metric heterogeneity that structural divergence exploits. This is a correlational observation—not a derivation from first principles—and measuring policy-space spread directly under MOEA/D versus diversity-preserving search is a target for future investigation. Random search, preserving maximum policy diversity, yields the largest contract benefit.

Table 3: Cohen's h for all 36 conditions ($n=20$/cell). Sem = Ad-hoc-Entropy vs. HEAS; Syn = Ad-hoc-Step vs. HEAS. KW*: $p \leq 0.05$ (optimizer effect). Bootstrap 95% CIs in artifact.

τ	Regime	Rand	NSGA	MD	KW p
0.05	Sem	1.25	1.36	1.26	0.016*
0.05	Syn	0.32	0.57	0.28	0.029*
0.10	Sem	0.96	0.99	0.86	0.355
0.10	Syn	0.20	0.30	0.35	0.045*
0.15	Sem	1.18	0.68	0.74	0.038*
0.15	Syn	0.58	0.41	0.30	0.578
0.20	Sem	0.90	0.99	0.46	0.001*
0.20	Syn	0.46	0.64	0.18	0.007*
0.25	Sem	0.86	0.56	0.26	0.004*
0.25	Syn	0.59	0.36	0.16	0.019*
0.30	Sem	0.67	0.60	0.30	0.009*
0.30	Syn	0.44	0.50	0.19	0.011*

Scope boundary, not algorithm-agnosticism. This result establishes a scope condition rather than an invariance claim: contract efficacy depends on optimizer-induced policy-space coverage. Pairing contracts with diversity-preserving search (NSGA-II, random) maximizes the divergence-prevention guarantee; decomposition-based optimizers (MOEA/D) attenuate but do not eliminate the benefit. The finding does *not* claim that contracts are optimizer-independent; it maps the boundary of the guarantee. Table 3 (Appendix B) reports full effect sizes with bootstrap 95% CIs for all 36 conditions.

5.8 Cross-Domain Portability

To complete the RQ2 cross-domain evidence, we ask whether the same framework code can be reused across domains. The answer is yes at the code level. The ecological and enterprise studies use the same optimization, tournament, and statistical infrastructure; only the domain-specific `metrics_episode()` implementations differ. In the enterprise tournament validation across 8 governance scenarios (20 repeats, 30 episodes per scenario), all four voting rules agree in 100% of cases and $P(\text{correct winner}) = 1.0$ at all tested budgets. This supports a narrow portability claim: the framework transfers across the tested ODE/tabular domains without rewriting framework logic. It does not yet establish portability to spatially-explicit or heterogeneous-agent ABMs.

5.9 Threats to Validity

Construct validity. Rank reversal rate may understate divergence (lower-ranked reversals uncounted); Pareto τ provides a complementary measure. The contract prevents structural inconsistency (different aggregation implementations) but not semantic inconsistency (incorrect but consistently applied `metrics_episode()`); `validate_metrics_episode()` catches type violations at development time.

Internal validity. Stage 1 ($n=15$) is underpowered ($\approx 47\%$ at $h=0.476$); results are exploratory. Stage 2 ($n=30$, pre-specified) is consistent in direction. Hyperparameters (pop=20, ngen=10, steps=150) are fixed across pathways, so any confound affects both equally. OOD validation (Exp. C) extends beyond training but only within the tested ecological family.

External validity. All case studies are parameter optimization of dynamical systems (acknowledged as intentional scope in Section 4). Spatially-explicit ABMs (Wilensky 1999; Collier and Ozik 2022) require topology-aware aggregation that the scalar contract does not yet support; this is a defined scope boundary, not an omission. Held-out scenarios outside the ecological training distribution are likewise a scope boundary.

6 CONCLUSION

Silent aggregation divergence is a hidden confound in ABM policy-search pipelines: when optimization, tournament, and inference stages independently implement metric extraction, champion selection reflects aggregation artifact rather than policy quality. HEAS addresses this by enforcing a single `metrics_episode()` callable at dispatch time—removing the confound by construction, not post-hoc reconciliation.

The controlled experiment (pre-specified, $n = 30$ Stage 2) confirms that removing the confound yields measurably better champion selection: HEAS reduces rank reversals by 50% relative to ad-hoc aggregation ($h = 0.215$; $p = 0.0067 < \alpha_{\text{conf}} = 0.025$, *confirmatory pass*; Stage 1 exploratory at $n = 15$, 47% power). The 32/32 held-out result is a null-safety check: ranking stability across held-out scenarios is evidence the contract removes the confound. The framework reduces coupling code by 97% and transfers across ecological, enterprise, and mean-field ODE domains. The multi-optimizer experiment establishes a scope condition: contract efficacy is optimizer-conditional, with diversity-preserving search (NSGA-II, random) yielding the largest guarantee and MOEA/D attenuating but not eliminating it.

Limitations and open questions. HEAS targets ODE/tabular dynamics, not spatial ABMs; its scalar contract (`dict[str, float]`) excludes non-scalar outputs such as agent-state distributions or spatial density maps—extending to these would require redesigning the tournament and inference interfaces. Metric semantics remain the modeller's responsibility. Open questions include contract performance in high-stochasticity regimes where episode variance dominates between-policy signal.

An anonymized artifact package containing all scripts and result files (S1–S10) is available for reviewers; upon acceptance, source code and raw data will be published with DOI.

A REPORTED MEANS AND SOURCES

Table 4: Key reported results with source file IDs.

Quantity	Value	Source
Coupling LOC reduction	97% (160 vs. 5)	S1
Rank reversal: HEAS vs. ad-hoc	0% vs. 6.7%, $h = 0.476$	S2
32-scenario champion wins	32/32, $p = 4.66 \times 10^{-10}$	S6b

Source Files. All experiment data and scripts (S1–S8, supplementary ablations) are provided in the artifact package under `experiments/results/`.

B AD-HOC-STEP BASELINE: REPRODUCIBILITY

Parameters. Pop = 20, ngen = 10, steps = 150; 8 scenarios, 100 episodes per scenario; 4 voting rules (argmax, majority, Borda, Copeland); $n = 15$ Stage 1 runs ($r = 0 \dots 14$), $n = 30$ Stage 2. Seeds: `seed = 42*1000 + run_id` (deterministic).

Per-stage aggregation (no shared registration). Optimizer reads `ep["final_biomass"]` (last step); Tournament reads `ep["mean_biomass"]` (episode mean); Inference reads `ep["roll10_biomass"]` (10-step rolling mean). Divergence accumulates silently.

Reversal rate. A run is a reversal if Ad-hoc-Step's tournament winner disagrees with HEAS's winner under *any* voting rule; ties broken by majority rule then alphabetical index. Stage 1: 1/15 runs reversed = 6.7% (Wilson 95% CI: [1.2%, 29.8%]).

REFERENCES

Andradóttir, S. 2015. "A Review of Random Search Methods". In *Handbook of Simulation Optimization*, edited by M. C. Fu, 277–292. New York, NY: Springer https://doi.org/10.1007/978-1-4939-1384-8_10.

Borshchev, A. 2013. *The Big Book of Simulation Modeling: Multimethod Modeling with AnyLogic 6*. Chicago, IL: AnyLogic North America.

Capri, S., M. Ignaccolo, G. Inturri, A. Pluchino, and A. Rapisarda. 2023. "Evaluating Transport Policies Through NetLogo Multi-Agent Simulation and Evolutionary Algorithms". *Transportation Research Procedia* 69:613–620.

DOI not yet indexed; Transportation Research Procedia, Vol. 69, pp. 613–620, 2023; Conference: AIIT 3rd International Conference on Transport Infrastructure and Systems (TIS ROMA 2022).

Collier, N., and J. Ozik. 2022. "Distributed Agent-Based Simulation with Repast4Py". In *Proceedings of the 2022 Winter Simulation Conference*, 192–206: IEEE https://doi.org/10.1109/WSC57314.2022.10015389.

Deb, K., A. Pratap, S. Agarwal, and T. Meyarivan. 2002. "A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II". *IEEE Transactions on Evolutionary Computation* 6(2):182–197 https://doi.org/10.1109/4235.996017.

Efron, B., and R. J. Tibshirani. 1994. *An Introduction to the Bootstrap*. New York: Chapman and Hall/CRC.

Fortin, F.-A., F.-M. De Rainville, M.-A. Gardner, M. Parizeau, and C. Gagné. 2012. "DEAP: Evolutionary Algorithms Made Easy". *Journal of Machine Learning Research* 13:2171–2175.

Fu, M. C. 2002. "Optimization for Simulation: Theory vs. Practice". *INFORMS Journal on Computing* 14(3):192–215 https://doi.org/10.1287/ijoc.14.3.192.113.

Hevner, A. R., S. T. March, J. Park, and S. Ram. 2004. "Design Science in Information Systems Research". *MIS Quarterly* 28(1):75–105 https://doi.org/10.2307/25148625.

Kazil, J., D. Masad, and A. Crooks. 2020. "Utilizing Python for Agent-Based Modeling: The Mesa Framework". In *Social, Cultural, and Behavioral Modeling (SBP-BRiMS)*, 308–317: Springer https://doi.org/10.1007/978-3-030-61255-9_30.

Kendall, M. G. 1938. "A New Measure of Rank Correlation". *Biometrika* 30(1–2):81–93 https://doi.org/10.1093/biomet/30.1-2.81.

Kwakkel, J. H. 2017. "The Exploratory Modeling Workbench: An Open Source Toolkit for Exploratory Modeling, Scenario Discovery, and (Multi-Objective) Robust Decision Making". *Environmental Modelling & Software* 96:239–250 https://doi.org/10.1016/j.envsoft.2017.06.054.

Liskov, B. H., and J. M. Wing. 1994. "A Behavioral Notion of Subtyping". *ACM Transactions on Programming Languages and Systems* 16(6):1811–1841 https://doi.org/10.1145/197320.197383.

Meyer, B. 1997. *Object-Oriented Software Construction*. 2nd ed. Prentice Hall.

Pangallo, M., T. Heinrich, and J. D. Farmer. 2019. "Best reply structure and equilibrium convergence in generic games". *Science Advances* 5(2):eaat1328 https://doi.org/10.1126/sciadv.aat1328.

Tekin, E., and I. Sabuncuoglu. 2004. "Simulation Optimization: A Comprehensive Review on Theory and Applications". *IIE Transactions* 36(11):1067–1081 https://doi.org/10.1080/07408170490500654.

Wilensky, Uri 1997. "NetLogo Wolf Sheep Predation Model". Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL. http://ccl.northwestern.edu/netlogo/models/WolfSheepPredation; Accessed March 2026.

Wilensky, Uri 1999. "NetLogo". Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL. http://ccl.northwestern.edu/netlogo/.

Zhang, Q., and H. Li. 2007. "MOEA/D: A Multiobjective Evolutionary Algorithm Based on Decomposition". *IEEE Transactions on Evolutionary Computation* 11(6):712–731 https://doi.org/10.1109/TEVC.2007.892759.

Zitzler, E., and L. Thiele. 1998. "Multiobjective Optimization Using Evolutionary Algorithms—A Comparative Case Study". In *Parallel Problem Solving from Nature — PPSN V*, 292–301: Springer https://doi.org/10.1007/BFb0056872.

AUTHOR BIOGRAPHIES

RUIYU ZHANG is a graduate student affiliated with the Department of Politics and Public Administration at The University of Hong Kong, Hong Kong SAR, China. Her e-mail address is Ruiyuzh@connect.hku.hk.

LIN NIE is an assistant professor affiliated with the Department of Applied Social Sciences at The Hong Kong Polytechnic University, Hong Kong SAR, China. His e-mail address is lin-apss.nie@polyu.edu.hk.

XIN ZHAO is a research student affiliated with the Department of Applied Social Sciences at The Hong Kong Polytechnic University, Hong Kong SAR, China. His e-mail address is xinnn.zhao@connect.polyu.hk.